This is exactly what happened in 2007 in the quant meltdown, and a year later for almost all HFs during the financial crisis.

3.5. A DEEPER LOOK AT VOLATILITY RISK

Given the myriad of HF investment styles, it might be surprising that many are economically identical in terms of selling volatility. The commonality is that many HFs choose strategies that most of the time deliver steady returns. But the cost of these steady returns during normal times is that when the bad times come, there are pronounced losses. Selling volatility is like selling hurricane insurance. You collect the premium most of the time. Then the hurricane hits and you are wiped out. The last financial hurricane occurred during 2008–2009.

A picture of a short volatility payoff is shown in Figure 17.7.

Figure 17.7 is the payoff of a short, out-of-the-money put.[34] Most of the time, the HF collects small and steady premiums equal to the price of the put. These profits look like "alpha" and HFs refer to this by various guises: "skill," "arbitrage," "mispricing," "market dislocation," "providing liquidity," "alternative beta," or just plain "short volatility." This premium does not come for free: there are occasional large losses when the assets fall sharply in price. In practice, the losses are higher than just simple put-selling because HFs use leverage. These losses do not show up

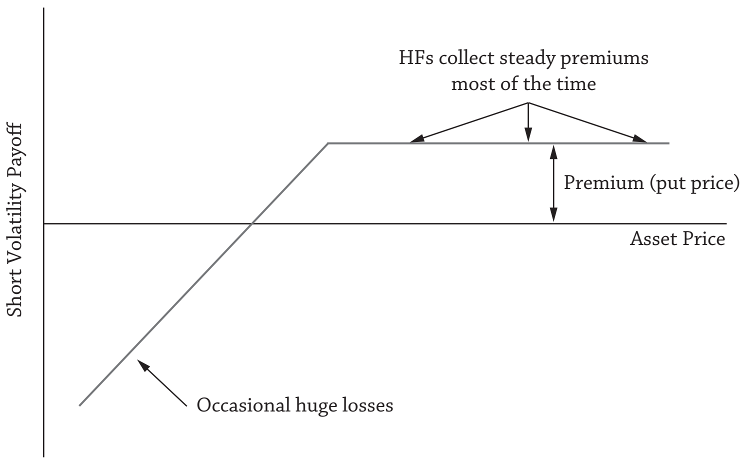

Figure 17.7

[34] This is actually the payoff of a rebalancing strategy as well. Any counter-cyclical investment strategy is short volatility, as chapter 4 shows. HFs just take this to the extreme. Lo (2001) creates an example of a HF that uses precisely this strategy to generate "alpha" and cheekily names the HF in his example "Capital Decimation Partners."